

Let $g \in \text{Aut}(\vec{T})$ s.t. $\forall v \in V$, if $(w, v) \in E\vec{T}$

$$g \cdot w = h_v \cdot w$$

Let $\{v_n\}_{n \in \mathbb{N}}$ enumeration of edges on the tree
with $v_0 = v$

Let $\{g_n\}_{n \in \mathbb{N}}$ sequence in $E_x(U, U)_{(v)}$

observe that $\vec{T} = \bigcup_{i \in \mathbb{N}} \text{inn}(v_i) \Rightarrow E_x(U, U)_{(v)} \leq \prod_{n \in \mathbb{N}} U$

is it closed?

no!

$W \curvearrowright \text{inn}(v)$ is same as $U \curvearrowright X$

G_x is open/compact $\forall x \in X$

$\Rightarrow G_x \cap U$ is open/compact in $U \Rightarrow \forall x \in X, |\text{orb}(x)| < \infty$

$w \in \text{inn}(v)$

$\Rightarrow$ in the same way $W \curvearrowright \text{inn}(w)$ is same as $U \curvearrowright X$

How get $E(U, U)_{(v)}$ compact with this?

$\{v_n\}_{n \in \mathbb{N}}$ enumeration of $V \setminus \vec{T}$, $\{g_n\}_{n \in \mathbb{N}}$ sequence in

$E(U, U)_{(v)} \Rightarrow$ let $N_1 \subseteq \mathbb{N}$ s.t. $g_n v_1 = w_1 \quad \forall n \in N_1$
 $N_2 \subseteq \mathbb{N}$ s.t. $g_n v_2 = w_2 \quad \forall n \in N_2$

Let $N' \subseteq N$ be, fixing $n_i \in N_i$,

$\{n_i\}_{i \in \mathbb{N}} \text{ s.t. } n_i \in N_i \text{ and } n_i \leq n_{i+1} \forall i \in \mathbb{N}$

$\Rightarrow g_{n_i} N_i = W_i$ and if $m > n_i$, $g_m N_i = W_i$

want now: $E_X(U, U)_{(w)}$ open in $E_X(G, U)$ and

$E_X(G, U)$ is t.d.s.c.l.c.

~~let $\emptyset \neq F \subseteq E_X(G, U)$ be the set of all finite permutations~~
~~t.d.s.c.l.c. comes from the $\text{Aut}(\bar{T})$ top.~~

s.c. comes from $\text{Aut}(\bar{T})$ top

$\hookrightarrow$ finite permutations make everything

t.d. comes from $\text{Aut}(\bar{T}) \leq \text{Sym}(N)$ with point-wise top

need: $E_X(G, U)$ is l.c.

~~because $E_X(G, U) \leq_c \text{Sym}(N)$ and every element~~

(compact open stab. on (G, x))

| $\text{stab}_{E_X(G, U)}(w) \cap W = \text{finite?}$

NO!!!

Want: show $E_X(U, U)_w \hookrightarrow E_X(G, U)$ is open
 $\Rightarrow$ 1 has compact neigh $\Rightarrow$ l.c.

let $\{g_n\}_{n \in \mathbb{N}} \subseteq E_X(G, U) \setminus E_X(U, U)_{(n)}$

s.t. $g_n \rightarrow g$

want: $g \in E_X(G, U) \setminus E_X(U, U)_{(n)}$

observe that if $g \in E_X(G, U)_{(n)} \Rightarrow \exists m \in \mathbb{N}$ s.t. $\forall m > n, g_m \neq g$

so, w.l.o.g., let $\{g_n\}_{n \in \mathbb{N}} \subseteq E_X(G, U)_{(n)} \setminus E_X(U, U)_{(n)}$

so for each $m \in \mathbb{N} \exists w \in \vec{T}$ s.t. $\sigma(g_m, w) \in G \setminus U$

suppose $g \in E_X(U, U)_{(n)}$

$\Rightarrow \sigma(g, w) \in U \quad \forall w \in \vec{T}$

$\Rightarrow \forall w \in \vec{T} \exists m \in \mathbb{N}$ s.t. $\forall m > n$

$\sigma(g_m, w) \in U$

because we have $\sigma(g, w) \in U \quad \forall w \in \vec{T}$

this would imply that eventually $g_n \in E_X(U, U)_{(n)}$



(Proof of Prop 9.2)

Prop 9.3

Suppose (G, X) is a t.d.l.c. s.c. group. Then

(a) if (G, X) is transitive and G is compactly generated, then $E_x(G, U)$ is compactly generated for any $U \in \mathcal{U}(G)$.

(b) If G is elementary, then $E_x(G, U)$ is elementary for any $U \in \mathcal{U}(G)$.

↳ Proof:

(a) $(\forall x, y \in X \exists g \text{ s.t. } x.g = y)$

$(\exists K \subseteq G, K \text{ compact, s.t. } G = \langle K \rangle)$

$(\nexists G \text{ is compactly generated, t.d.l.c. Polish} \Rightarrow \exists A \subseteq_{\text{finite}} G \text{ s.t. } G = \langle A \rangle U, U \in \mathcal{U}(G))$

Let $U \in \mathcal{U}(G)$ and $A \subseteq_{\text{fin}} G$ s.t. $\langle A \rangle U = G$

Let $v \in V\vec{T} \Rightarrow E_x(U, U)_{(v)}$ is compact

want: A finite set s.t. $\langle A \rangle E_x(U, U)_{(v)} = G$

Let δ be bi-ray s.t. $v \in \delta$ and it is ~~closed~~ monoch.

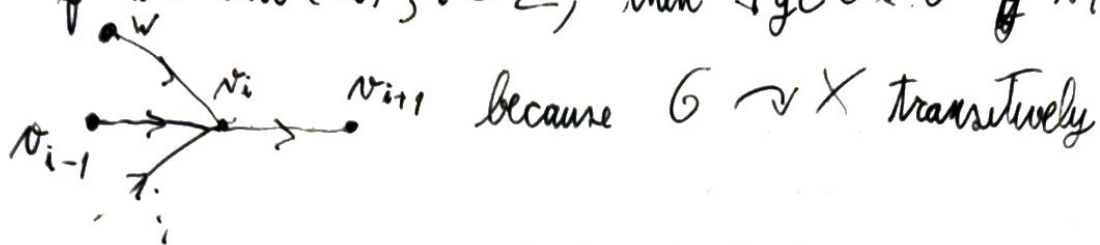
Let $\eta(v) = 0$

Let a be $a(v) = v_0$
 $a^{-1}(v) = v_{-1}$

Let $B = \langle A \cup \{a\} \rangle \Rightarrow \langle B \rangle E_x(U, U)_{(v)} \curvearrowright \text{inn}(v)$

$G \xrightarrow{\text{''}} X$ (because $\langle A \rangle U = G$)

if $w \in \text{inn}(v_i)$, $i \in \mathbb{Z}$, then $\exists g \in G$ s.t. $v_{i-1}g = w$



$$\Rightarrow v_0 a^{i-1} g = w$$

and inductively one can see we can send v_0 to $w \in V$

$\Rightarrow G = \langle B \rangle E_X(V, V)_{(v_0)} \Rightarrow$ it is compactly generated

(ii) Let G elementary

$\Rightarrow \exists (\mathbb{Q}_n)_{n \in \mathbb{N}}$ sequence of compactly generated subgroups s.t. $\bigcup_{n \in \mathbb{N}} \mathbb{Q}_n = G$ and $\mathcal{G}(\mathbb{Q}_n) \leq w$

~~$\mathbb{Q}_n = \langle \mathbb{Q}_n \rangle$~~

~~we can assume~~

~~$\mathbb{Q}_n = \langle \mathbb{Q}_n \rangle$~~

~~Let $S_w = \bigcup_{n \in \mathbb{N}} S_n$ (all finite permutations) with discrete top~~

~~S_w is elementary~~

~~$\Rightarrow \mathbb{Q}_n \times S_n$ is a sequence of compactly gen subgroups that~~

~~goes to S_w (which is elementary)~~

~~and S_w is transitive on X~~

Let $\{x_i\}_{i \in \mathbb{Z}} = X$ and define $x_i \cdot g = x_{i+1}$, $x_i \cdot g^{-1} = x_{i-1}$

$\Rightarrow \langle g \rangle \curvearrowright X$ is transitive and $\langle g \rangle \cong \mathbb{Z}$

put the discrete top on $\langle g \rangle$ and let

$G \times \mathbb{Z} \Rightarrow Q_n \times \mathbb{Z}$ are all compactly generated and elementary

and $G \times \mathbb{Z} = \bigcup_{n \in \mathbb{N}} Q_n \times \mathbb{Z}$
 $\hookrightarrow$ all transitive and comp gen.

$\Rightarrow$ let $U \in \mathcal{U}(G)$ and, w. l. o. g., let's assume $U \leq Q_n \ \forall n \in \mathbb{N}$

$\Rightarrow E_x(Q_n \times \mathbb{Z}, U)$ is compactly generated, for all $n \in \mathbb{N}$

$$\bigcup_{n \in \mathbb{N}} E_x(Q_n \times \mathbb{Z}, U_x) = E_x(G \times \mathbb{Z}, U_x)$$

if $g \in E_x(G \times \mathbb{Z}, U_x)$

for all $v \in T\vec{V}$
 $\Rightarrow \sigma(g, v) \in G \times \mathbb{Z} \setminus U_x$ for only finitely many $v \in T\vec{V}$

let $v_1, \dots, v_m$ be s.t. $\sigma(g, v_i) \in G \times \mathbb{Z} \setminus U_x$

$\Rightarrow \exists m \in \mathbb{N}$ s.t. $\sigma(g, v_i) \in Q_m \times \mathbb{Z} \setminus U_x, \forall i \in \{1, \dots, m\}$

$\Rightarrow g \in \bigcup_{n \in \mathbb{N}} E_x(Q_n \times \mathbb{Z}, U_x)$

Ex: $E_x(G, U) \leq_c E_x(G \times \mathbb{Z}, U \times \{1\})$
 $\forall G$ ^{Top} group

Let g_n $n \in \mathbb{N}$ sequence in $E_x(G, U)$ s.t. $g_n \rightarrow g$
 $\Rightarrow g \in E_x(G, U)$

Ex: $E_x(\mathbb{Q}_n \times \mathbb{Z}, U \times \{1\})$ is elementary $\forall n \in \mathbb{N}$

Let $\mathcal{G}(\mathbb{Q}_n \times \mathbb{Z}) = \alpha$, $\alpha < \omega_1$ ~~AAAAAAAAAAAAAAAAAAAAAAAAAAAA~~

Ex: If G is compgen/trans $\Rightarrow E_x(G, U)$ is elementary

Let $\mathcal{G}(G) = \alpha$, $\alpha < \omega_1$

Induction on α ? suppose it work $\forall \beta \leq \alpha$ and let

$\mathcal{G}(G) = \alpha + 1$

$\hookrightarrow$ hard to see base case ...?

Case $G = \{1\}$ $\mathcal{G}(G) = 1 \Rightarrow G = \{1\}$

$\Rightarrow G \times \mathbb{Z} \cong \mathbb{Z}$

$E_x(\mathbb{Z}, \{1\}) \cong F_2$ with discrete top

a' go back at ^{to} ~~the color~~ ~~the ray~~ and a move towards the ray

b moves the color $\uparrow$ and b^{-1} $\downarrow$



$$\Rightarrow \mathcal{G}(E_x(6 \times \mathbb{Z}, U_x \{1\}) = 2$$

$$\text{Suppose } \mathcal{G}(Q_1) \nmid \beta \leq \alpha, \mathcal{G}(6) = \beta \Rightarrow \mathcal{G}(E_x(6 \times \mathbb{Z}, U_x \{1\})) < \omega_1$$

$\forall U \in \mathcal{U}(6)$

Case 1: ~~is limit ordinal successor of~~ $\alpha = \beta + 1$

$$\text{let } \mathcal{G}(6) = \alpha + 1$$

Case 1: $\alpha + 1 = \beta + 2$

$$\mathcal{G}(\text{Res}_2(6)) = \beta + 1 \text{ and}$$

$$\mathcal{G}(E_x(\text{Res}_2(6) \times \mathbb{Z}, U_x \{1\}))$$

$$\mathcal{G}(E_x(6 \times \mathbb{Z}, U_x \{1\})) = \sup_n \mathcal{G}(E_x(Q_n \times \mathbb{Z}, U_x \{1\})) + 1$$

$$\text{but } \mathcal{G}(Q_n) \leq \alpha \xrightarrow{\text{by ind.}} \sup_n \mathcal{G}(E_x(Q_n \times \mathbb{Z}, U_x \{1\})) < \omega_1$$

$$\Rightarrow \sup_n \mathcal{G}(E_x(Q_n \times \mathbb{Z}, U_x \{1\})) + 1 < \omega_1$$



$$P_x(G, U) = \{g \in L_x(G, U), \eta(gv) = \eta(v) \forall v \in VT\}$$

$\Rightarrow$ This subgroup is normal, open and $E_x(G, U)/P_x(G, U) \cong \mathbb{Z}$

Def: This subgroup is normal because $\forall h \in P_x(G, U)$

$$gv = v \Rightarrow hgh^{-1}v = v \Rightarrow hgh^{-1} \in P_x(G, U)$$

$\Rightarrow$ Let $\{g_n\}_{n \in \mathbb{N}}$ be sequence with g_n outside $P_x(G, U)$

$$\Rightarrow \forall v \in VT \forall n \in \mathbb{N}, g_nv \neq v \text{ and } \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N$$

$$g_nv = gv \Rightarrow gnv \neq v \forall v \in VT \Rightarrow g \notin P_x(G, U)$$

$\Rightarrow E_x(G, U) \setminus P_x(G, U)$ is open closed

$\rightarrow$ Define $\alpha \in E_x(G, U)$ as:

Let δ be a monochromatic ray with $v_0 \in \delta$ be s.t.

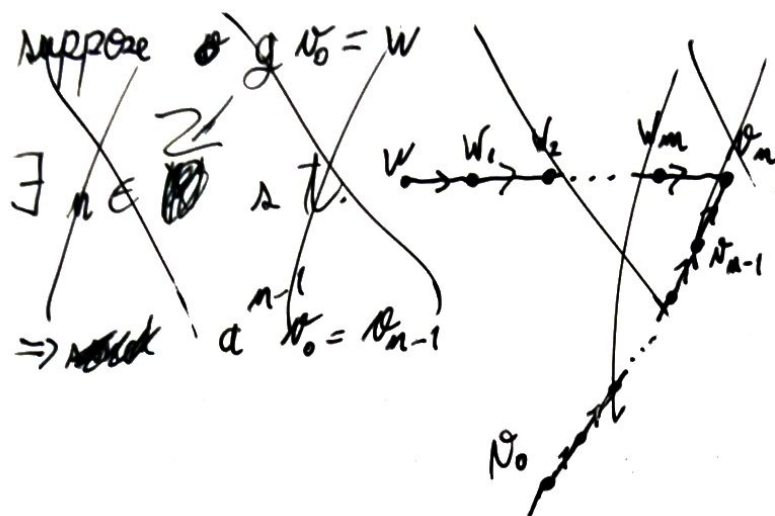
$$\eta(v_0) = 0$$

$\alpha^n v_0 = v_n$ the element at the ray with $\eta(v_n) = n$
($n \in \mathbb{Z}$)

if $w \notin \delta \Rightarrow$ we want w be s.t. $\eta(\alpha^n w) = \eta(w)$
the color of w and $\alpha^n w$ is the same.

one can easily see that $\langle a \rangle \cap P_x(6, U) = \{1\}$

Let $g \in F_x(6, U)$. If: $g = a^{n_1} h_1 a^{n_2} h_2 \dots a^{n_m} h_m$
with $n_i \in \mathbb{Z}$, $h_i \in P_x(6, U)$



~~$\exists h_i \in P_x(6, U)$ s.t. $h_i(v_{n-1} \Rightarrow v_n)$ and $h_i v_{n-1} = v_n$~~

Define $d(v, w) = \min \{e_1, \dots, e_n\}$ s.t. $e_i = (v, v_i)$ or (v_i, v)
and $e_n = (v_{n-1}, w)$ or (w, v_{n-1})

if $d(v, w) = 1 \Rightarrow w = v_1$ or $w \in \text{inn}(v)$

if $w \in \text{inn}(v) \Rightarrow a^{-1} v_0 \in \text{inn}(v) \Rightarrow \exists h_1 \in P_x(6, U)$

s.t. $h_1 a^{-1} v_0 = w$ and $h_1 v_0 = v_0$

Why?

$g v_0 = w = g a v_{-1}$ and $g a v_0 \neq v_0$

$g a v_0 = g v_1 \in \text{inn}(v_1) \Rightarrow \eta(g a v_0) = \eta(v_0)$
 $\Rightarrow g a \in P_x(6, U)$

$g: \exists h_2 \in P_x(G, U) \text{ s.t.}$

$$h_2 h_1 a^{-1} = g$$

$$\text{let } w_1 \in \sqrt{I} \Rightarrow g w_1 = w_2$$

lot of work but I get the idea now.

Prop: (G, X) transitive t.d.l.c.s.c. permutation grp, Then $E_x(G, U)$ is monolithic, and the monolith is:

$$M := \overline{[P_x(G, U); P_x(G, U)]}.$$

If G is also top. perfect, then $P_x(G, U)$ is the monolith.

$\hookrightarrow$ $g: \text{let } N \trianglelefteq \overline{E_x(G, U)}$. Want:

$$\overline{[P_x(G, U); P_x(G, U)]} \leq N$$

$$\text{let } g \in \overline{[P_x(G, U); P_x(G, U)]} \Rightarrow \exists \{g_n^{(1)}, g_n^{(2)}\}_{n \in \mathbb{N}} \subseteq [P_x(G, U); P_x(G, U)]$$

$$\text{s.t. } g_n \rightarrow g \quad [g_n^{(1)}, g_n^{(2)}] \xrightarrow{n \rightarrow \infty} g$$

$\hookrightarrow$ normal $\Rightarrow$ if $h_1 \in N$, $g \in E_x(G, U) \Rightarrow \exists h_2 \in N$ s.t.

$$g h_1 g^{-1} = h_2 \Rightarrow \eta(h_2 \sigma) = \eta(g h_1 g^{-1})$$

$$\forall \sigma \in \sqrt{I}$$

let ~~$v_1, v_2 \in \text{inn}(U)$~~

~~$\Rightarrow \exists g \in G \text{ s.t. } g v_1 = v_2$~~

let ~~$h \in N$ and $g \in G$~~

~~$\sigma(h, v_1) = g \sigma(h, v_1) g^{-1}$~~

~~$\Rightarrow N \trianglelefteq \text{inn}(U)$ is normal in G~~

~~N~~

let $[P_x(G, U), P_x(G, U)] \ni [g_1, g_2]$

$g_1 g_2 g_1^{-1} g_2^{-1} \in N?$

$\hookrightarrow \exists v \in V^{\neq} \text{ s.t. } g_1 g_2 g_1^{-1} g_2^{-1} v = v? \quad (\eta(v) = \eta(g_1 g_2 g_1^{-1} g_2^{-1} v))$

Prop: For (G, X) a t.d.l.c.s.c. ^{permutation} grp, each horoball factor K/L admits a $P_x(G, U)$ -equivariant isomorphism $\psi: K/L \rightarrow \bigoplus_{\partial X} (G, U)$ where X is the horoball corresponding to K .

(Obs: Let $H_i = \text{Stab}_{P_x(G, U)}(X_i)$ of a horoball $X_i \Rightarrow H_i \trianglelefteq P_x(G, U)$, and $H_i \leq H_j$ for $i \leq j$. We thus have canonical normal factors H_{i+1}/H_i of $P_x(G, U)$ (horoball factors))

$\hookrightarrow$ Def: $(X_i = \{v \in V\bar{T} : \eta(v) \geq i\}$

No need

Prop: Suppose (G, X) is a transitive permutation t.d.l.c.s.c. group and $U \in \mathcal{U}(G)$. If G is top. simple and compactly generated, then $P_x(G, U)$ is a stacking type chief factor of $P_x(G, U)$, and every chief factor of $P_x(G, U)$ is a horoball factor.

$\hookrightarrow$ Stacking type if $B_0^{\min} \neq \emptyset$ and for all $a, b \in B_0^{\min}$,

$\exists \psi \in \text{Aut}(G)$ s.t. $\psi.a \leq b$

$\hookrightarrow B_0^{\min}$ is the set of minimally covered chief blocks

$\{ (a \in B_0 \text{ s.t. exists } N \leq G \text{ covering the block})$

$\hookrightarrow N/M$ covers a if $C_G(a)$ contains M but not N .

B_0 is the set of chief factor blocks